

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 45

WORLD CLIMATIC TYPES
THE ARCTIC CLIMATE

WORLD CLIMATIC TYPES

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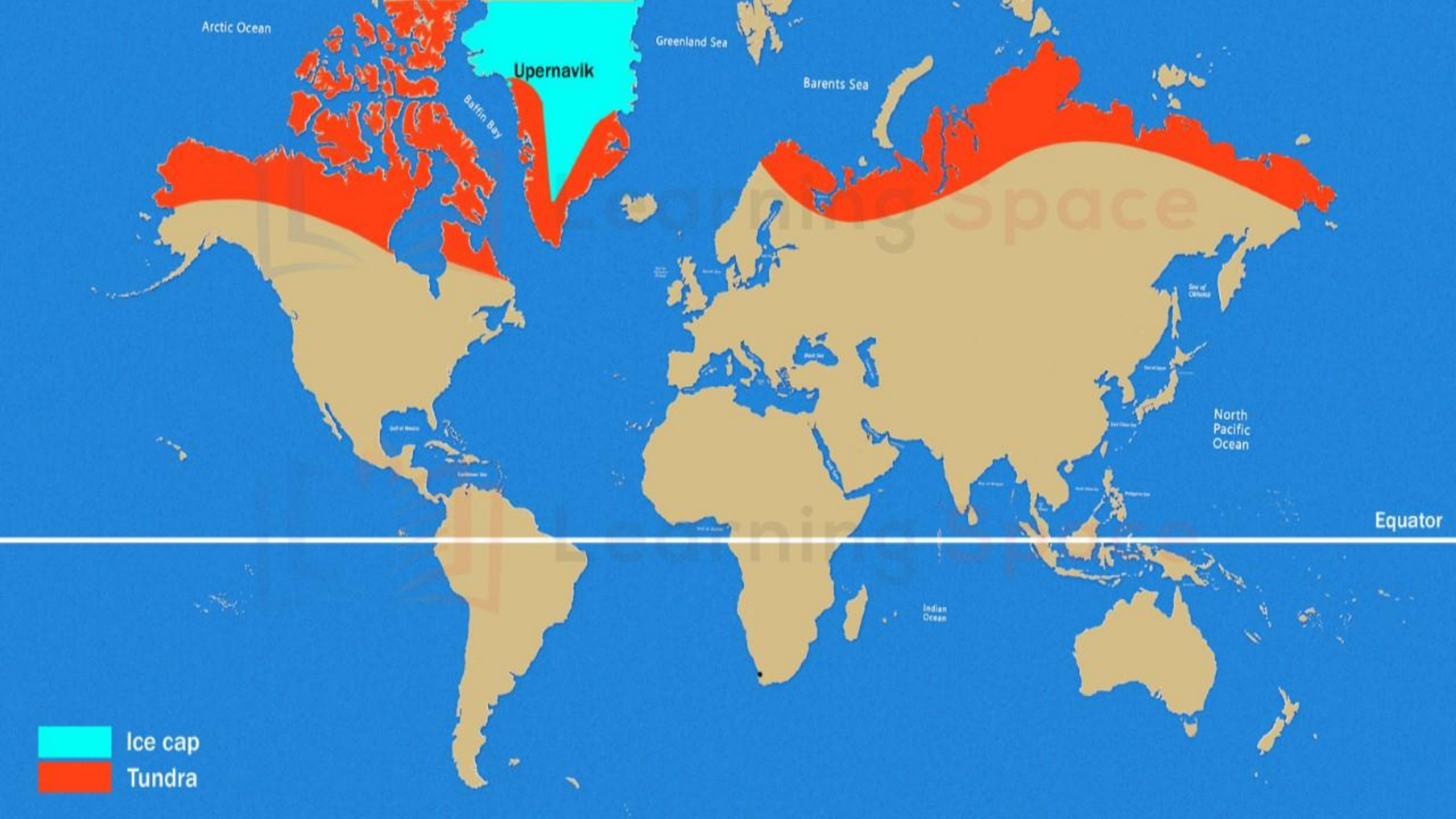
THE ARCTIC OR POLAR CLIMATE

DISTRIBUTION



1 The polar type of climate and vegetation is found mainly north of the Arctic Circle in the northern hemisphere.

2 The ice-caps are confined to Greenland and to the Highlands of these high-latitude regions, where the ground is permanently snow-covered.



WORLD CLIMATIC TYPES

9 THE ARCTIC OR POLAR CLIMATE DISTRIBUTION



3 The low-lands, with a few months ice-free, have tundra vegetation. They include the coastal strip of Greenland, the barren grounds of northern Canada and Alaska and the Arctic seaboard of Eurasia.

4 In the southern Hemisphere, the virtually uninhabited continent of Antarctica is the greatest single stretch of ice-cap where the layers of permanent ice are as thick as 10,000 feet.

9 THE ARCTIC CLIMATE

CLIMATE

a TEMPERATURE



1 The polar climate is characterized by a **very low mean annual temperature** and its warmest month in June seldom rises to more than 50°F/ 10°C.

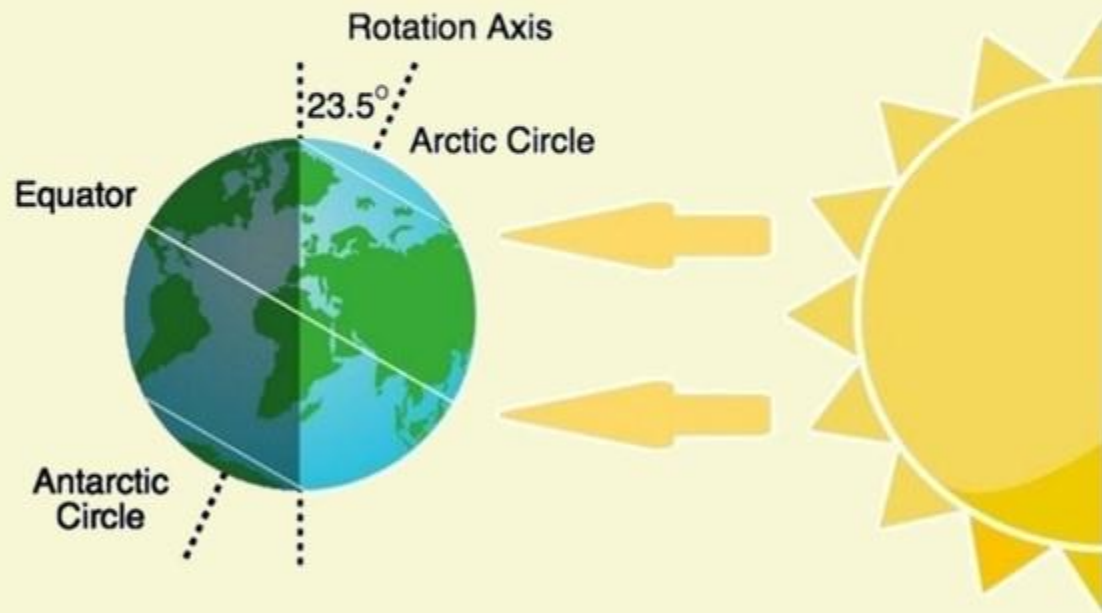
2 In mid-winter (January) temperatures are as low as -35°F and much colder in the interior.

3 Normally, **not more than four months have temperatures above freezing-point!** Winters are long and very severe; summers are cool and brief.

9 THE ARCTIC CLIMATE

CLIMATE

a TEMPERATURE



4 Within the Arctic and Antarctic Circles, there are weeks of continuous darkness.

5 At the North Pole, there are six months without light in winter.

6 Despite the long duration of sunshine in summer, when the sun does not set, temperatures remain low because the sun is low in the sky and much of the warmth of its faint rays is either reflected by the ground snow, or used up in melting the ice.

9 THE ARCTIC CLIMATE

CLIMATE

a

TEMPERATURE



7 It has little power left to raise the air temperature

8 Frost occurs at any time and blizzards reaching a velocity of 140 miles an hour are not infrequent.

9 They can be very hazardous for the polar inhabitants. In coastal districts, where warmer water meets cold land, thick fogs may develop.

10 They last for days, and in many instances it is not possible to see for more than a few feet.

9 THE ARCTIC CLIMATE

CLIMATE

b PRECIPITATION



Blizzards in Arctic

1 Precipitation is mainly in the form of snow falling in winter and being drifted about during blizzards.

2 Snowfall varies with locality; it may fall either as **ice crystals** or large, **amalgamated snowflakes**.

3 **Convectional rainfall** is generally **absent** because of the low rate of evaporation and the lack of moisture in the cold polar air.

9 THE ARCTIC CLIMATE

NATURAL VEGETATION



- 1 In such an adverse environment as the tundra, few plants survive.
- 2 The greatest inhibiting factor is the region's deficiency in heat.
- 3 With a growing season of less than three months and with the very low temperatures and less sunshine, it can be said that there are no large trees in the tundra.

9 THE ARCTIC CLIMATE

NATURAL VEGETATION



Mosses

- 4 Such an environment can support only the lowest form of vegetation- **mosses, lichens, and sedges**.
- 5 Drainage in the tundra is usually poor as the **sub-soil is permanently frozen**.
- 6 Ponds and marshes and waterlogged areas are found in hollows.
- 7 In the more sheltered spots, **stunted birches, dwarf willows and undersized alders** struggle for a meagre existence.

9 THE ARCTIC CLIMATE

NATURAL VEGETATION



8 Climatic conditions along the coastal lowlands are a little more favorable.

9 Here are found some hardy grasses and the reindeer moss which provide the only pasturage for the herbivorous animals like reindeer.

10 In the brief summer, when the snows melt and the days are warmer and longer, berry-bearing bushes and Arctic flowers bloom.

9 THE ARCTIC CLIMATE

NATURAL VEGETATION



11 Though short-lived, they brighten the monotonous tundra landscape into “Arctic Prairies”.

12 In the summer, the tundra is full of activities.

13 Birds migrate north to prey on the numerous insects which emerge when the snow thaws.

14 Mammals like the wolves, foxes, musk-ox, Arctic hare and lemmings also live in tundra regions.

9 THE ARCTIC CLIMATE

HUMAN ACTIVITIES



- 1 Human activities of the tundra are **largely confined to the coast**. Where plateau and mountains increase the altitude, it is uninhabitable, for these are **permanently snow-covered**.
- 2 The few people who live in the tundra live a **semi-nomadic life** and have to adapt themselves to the harsh environment.
- 3 In **Greenland, northern Canada and Alaska** live the **Eskimos**, numbering less than 28,000 today.

9 THE ARCTIC CLIMATE

HUMAN ACTIVITIES



4 They used to live as hunters, fishers and food-gatherers but in recent years more and more of them are settling in permanent huts.

5 The seasonal changes in climate necessitate a nomadic way of life. During winter they live in compact igloos and in summer when they move out to hunt they pitch portable tents of skins by the side of streams.

6 Their food is derived from fish, seals, walruses and polar bears.

9 THE ARCTIC CLIMATE

HUMAN ACTIVITIES



Harpoons & Kayaks

- 7 In the last fifty years through their contact with the Europeans, the way of life of the Eskimos has undergone tremendous changes.
- 8 Coastal villages have permanent wooden houses complete with modern facilities; speed-boats are replacing frail kayaks.
- 9 Deadly rifles instead of traditional harpoons are used to track down animals and seals.

9 THE ARCTIC CLIMATE

HUMAN ACTIVITIES



**Genetically mutated foxes
in fur farms**

10 Fur-bearing animals are being reared on a commercial scale and fishing, too, is being commercialized.

11 In some more accessible parts of Canada and Alaska, schools have been established and the Eskimo children are being taught the skills which will allow them to fit into the modern way of life.

9 THE ARCTIC CLIMATE

HUMAN ACTIVITIES

- 12** In the Eurasian tundra are other nomadic tribes such as the **Lapps** of northern Finland and Scandinavia, the **Samoyeds** of Siberia (from the Ural Mountains and the Yenisey basin), the **Yakuts** from the Lena basin, and **Chuckchi** of Chukotka peninsula.



Lapps/Saami of Scandinavia



Samoyed tribe

9 THE ARCTIC CLIMATE

HUMAN ACTIVITIES



Chukchi sea



The Chukchi people of the Chukotka Sea, the nation of reindeer herders and whale hunters

9 THE ARCTIC CLIMATE

HUMAN ACTIVITIES



13 They wander with their herds of reindeer across the Eurasian tundra where there are pastures. Many of them have taken to a more settled life.

14 In the former Soviet Union regions, large farms have been established for raising reindeer and for breeding fur-bearing animals.

9 THE ARCTIC CLIMATE

THE IMPORTANCE AND RECENT DEVELOPMENT OF THE ARCTIC REGION



- 1 The Arctic region, once regarded as completely useless, is now of some economic importance.
- 2 Apart from the efforts of the various Governments in assisting the advancement of the Arctic inhabitants the Eskimos, Lapps, Samoyeds etc. **new settlements have sprung up because of the discovery of minerals.**
- 3 Gold is mined in Alaska, nickel near Petsamo USSR, Petroleum in the Kenai Peninsula, Alaska and copper at the Rankin Inlet, Canada.

9 THE ARCTIC CLIMATE

THE IMPORTANCE AND RECENT DEVELOPMENT OF THE ARCTIC REGION



4 Coal has been mined in Spitsbergen for a long time and also in Alaska.

5 With the declining reserves of iron ore around Lake Superior, the Great Lakes industrial concerns re using more and more iron from large iron ore deposits in Labrador.

6 New railway line have been constructed to bring the ores to the St. Lawrence River for subsequent shipment to the major industrial districts.

9 THE ARCTIC CLIMATE

THE IMPORTANCE AND RECENT DEVELOPMENT OF THE ARCTIC REGION



Ice breaker ships

- 7 Rich deposits of iron ores at Kiruna and Gallivare in Sweden have made it possible for Sweden to enjoy a prosperous export trade in iron and steel and other metallurgical products.
- 8 With the establishment of ports on the Arctic seaboard of Eurasia, it is now possible to ship timber and fur from Siberia.
- 9 Though the ports, such as Igarka at the mouth of the Yenisey, are not ice-free, modern ice-breakers keep the passage open most of the time.

9 THE ARCTIC CLIMATE

THE IMPORTANCE AND RECENT DEVELOPMENT OF THE ARCTIC REGION



10 The healthy air and its preservative qualities (it is practically germ-free) are important factors for human research activities.

11 Scientists, meteorologists and explorers have lived in the Arctic and Antarctica, making studies of their geology, weather conditions, plant and animal life, that will be of great significance in years to come.

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For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao

K. Srikanth

D. Sunil Kumar

L. V. Krishna

Y. Deepthi

Amrita Naidu

M. Binukrishna

G. Ravi Babu

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